

Junhong Cheng

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Education

Shanghai Jiao Tong University

Sep. 2023 – Jun. 2027 (expected)

B.S. in Computer Science, **ACM Honors Class**

GPA: **90.0/100**

Selected coursework: Mathematical Logic (A+, 100/100); Principles and Practice of Computer Algorithms (A+, 97/100); Compiler Design (A, 94/100); Computer Architecture; Operating Systems; Machine Learning Systems; Model Checking.

Research Interests

My research interests lie at the intersection of systems security and performance, with a focus on **microarchitectural security**, speculative execution defenses, and **trusted execution environments (TEEs)**. I am also interested in **AI safety** and **agent safety**, particularly in applying systems security principles to build trustworthy AI systems and agents.

Research Experience

DCASL, University of California San Diego

May 2026 – present

Advisor: Prof. Dean Tullsen; collaborators: Luyi Li and Hosein Yavarzadeh

- **Intel Defense Profiling:** Reverse engineer Intel defense mechanisms, measure their overhead, and explore lower-latency alternatives.

Network Security and Privacy Protection (NSEC) Lab, SJTU

Jun. 2025 – Jun. 2026

Advisor: Prof. Guoxing Chen

- Second author of **Attest the Whole, Verify Incrementally** — **CCS 2026: Accepted**.
- Worked on reducing enclave cold-start latency by dynamically loading enclave pages after startup.

Personal Projects

- **Mx Compiler** (Java): Compiler for a C-like educational language; implemented Mem2Reg, ADCE, and SSA-based register allocation, with performance comparable to -O2 in the course evaluation.
- **Verilog CPU**: Implemented Tomasulo scheduling and support for basic RISC-V integer instructions.
- **Raft** (Go): Implemented replicated-log consensus for an algorithms course project, addressing network delays, partitions, packet loss, duplication, and reordering.

Awards and Honors

Zhiyuan Honors Scholarship, 2023 and 2024

Top 2% at SJTU

Teaching Assistant Experience

Great Ideas in Computer Science

Fall 2024

Instructor: Alei Liang

Principles and Practice of Computer Algorithms

Summer 2025

Instructor: Yong Yu

Student Leadership

Class Monitor, ACM Honors Class of 2023

Sep. 2024 – Sep. 2025

Organization Committee Member, ACM Honors Class

Sep. 2025 – Sep. 2026

Skills

Programming: C++, Go, Java, Verilog **Tools:** Git, Markdown, $\text{\LaTeX}$

Languages: Chinese (native), English (fluent); TOEFL: 110/120 (5.5/6.0), Speaking: 5.0/6.0.